

Nmap TCP & UDP Scans

Nmap (Network Mapper) is a popular network scanning tool used to discover hosts, open ports, and services running on a target system. TCP and UDP scans help identify which ports are open and what services are available on a target machine.

TCP SYN Scan (-sS)

A TCP SYN Scan, also known as a Stealth Scan, sends a SYN packet to the target port without completing the TCP handshake.

Example: `nmap -sS 192.168.56.101.`

Purpose: Detect open TCP ports, perform fast scanning, and reduce logging compared to a full TCP connect scan.

UDP Scan (-sU)

A UDP Scan checks for services running on UDP ports.

Example: `nmap -sU 192.168.56.101.`

Purpose: Identify open UDP services such as DNS, SNMP, and DHCP and discover additional attack surfaces.

Service Version Detection (-sV)

Service Version Detection identifies the software and version running on open ports.

Example: `nmap -sV 192.168.56.101.`

Purpose: Detect service versions, identify software running on ports, and discover outdated or vulnerable services.

Operating System Detection (-O)

Operating System Detection attempts to identify the operating system running on the target machine.

Example: `nmap -O 192.168.56.101.`

Purpose: Identify the target operating system, determine device type, and improve vulnerability assessment accuracy.

Creating a Scan Report

After completing Nmap scans, the results should be documented in a scan report. Include the target IP address, date and time of scan, open TCP and UDP ports, running services, service versions, operating system details, and observations.

Summary:

TCP SYN Scan (-sS) identifies open TCP ports. UDP Scan (-sU) identifies open UDP services. Service Version Detection (-sV) detects software versions. OS Detection (-O) identifies the operating system. A scan report documents findings and observations.

The results can be seen in my screenshots.